

Control work 3, and work on the mistakes

Rational numbers in one paper, and the first quarter closed.

LESSON 44–45

2 × 40 MIN

MATEMATIKA 7, NAZORAT ISHI 3

S8 7 REVIEW

LESSON CLOCK

3

40'

12'

20'

5'

Instructions	3 min
The paper	40 min
Answers	12 min
Diagnosis and rewrite	20 min
The quarter	5 min

LEARNING OBJECTIVES

- Convert between fractions, decimals and mixed numbers under time.
- Add and subtract rational numbers with signs.
- Solve a simple equation in $\mathbb{Q}$.
- Classify each lost mark and rewrite the whole solution.

TEXTBOOK

National	pp. 95–117
Cambridge	Stage 8, pp. 66–76
Workbook	Control paper M3

01

Explanation

The paper — 30 marks, 40 minutes

Q	TASK	MARKS	FROM
1	Write -2.4 and $3\frac{1}{4}$ as fractions $\frac{p}{q}$	4	L37–38
2	Convert $\frac{7}{8}$ and $\frac{5}{12}$ to decimals	4	L37–38
3	Compute $\frac{2}{3} + \frac{3}{4}$ and $-\frac{1}{2} + \frac{1}{5}$	6	L39–40
4	Compute $\frac{5}{6} - \frac{3}{8}$ and $-\frac{2}{5} - \frac{1}{2}$	6	L41–42
5	Compute $3\frac{1}{3} - 1\frac{3}{4}$	5	L41–42
6	Solve $x + \frac{2}{3} = \frac{1}{6}$	5	L43

WHERE THE MARKS ACTUALLY GO

Q1 carries one mark for simplifying; Q2 one for showing the recurrence; Q3 two for the LCM; Q5 two for converting to improper fractions; Q6 one for the check.

Naming the slip

SLIP	WHAT IT LOOKS LIKE	THE FIX
fraction not simplified	$-\frac{24}{10}$	$-\frac{12}{5}$
recurrence not shown	0.41	$0.41666\dots$
denominators added	$\frac{2}{3} + \frac{3}{4} = \frac{5}{7}$	$\frac{17}{12}$
wrong sign kept	$-\frac{1}{2} + \frac{1}{5} = \frac{3}{10}$	$-\frac{3}{10}$
mixed number sign	$-2\frac{1}{3} = -2 + \frac{1}{3}$	$-\frac{7}{3}$
borrowing wrong	$3\frac{1}{3} = 2\frac{11}{3}$	$2\frac{4}{3}$
equation solved by adding	$x = \frac{1}{6} + \frac{2}{3}$	$x = \frac{1}{6} - \frac{2}{3}$

Name the slip in the margin, then rewrite the whole solution — not the wrong line.

The quarter in one page

BLOCK	THE SENTENCE
the basic figures	point, line and plane, and the axioms that govern them
segments and angles	both add when a point or a ray lies inside
perpendicular lines	adjacent add to 180° , vertical are equal
the integers	invented so that subtraction always works
the sign rules	alike gives +, unlike gives –
the rationals	invented so that division always works
fractions	a common denominator, then the numerators

LOOKING FORWARD

Quarter II returns to geometry for the congruence of triangles, then opens algebra proper with powers, monomials and polynomials. The sign rules of this quarter are used in every line of it.

02 Worked examples

EXAMPLE 1 Model answer, Q3: $\frac{2}{3} + \frac{3}{4}$ and $-\frac{1}{2} + \frac{1}{5}$.

① LCM of 3 and 4 is 12.

② $\frac{8}{12} + \frac{9}{12} = \frac{17}{12} = 1\frac{5}{12}$

③ LCM of 2 and 5 is 10.

④ $-\frac{5}{10} + \frac{2}{10} = -\frac{3}{10}$

The half is bigger.

ANSWER

$1\frac{5}{12}$ and $-\frac{3}{10}$

EXAMPLE 2 Model answer, Q5: $3\frac{1}{3} - 1\frac{3}{4}$.

① Convert: $\frac{10}{3} - \frac{7}{4}$.

② LCM 12: $\frac{40}{12} - \frac{21}{12}$.

③ $= \frac{19}{12}$

④ $= 1\frac{7}{12}$

ANSWER

$$1\frac{7}{12}$$

EXAMPLE 3 Model answer, Q6: solve $x + \frac{2}{3} = \frac{1}{6}$.

① $x = \frac{1}{6} - \frac{2}{3}$

Subtract, do not add.

② $= \frac{1}{6} - \frac{4}{6}$

③ $= -\frac{3}{6} = -\frac{1}{2}$

④ Check: $-\frac{1}{2} + \frac{2}{3} = \frac{1}{6}$ ✓

ANSWER

$$x = -\frac{1}{2}$$

03 Interactive model

Mark Q3 aloud with a paper strip folded into twelfths; the wrong answer of five sevenths becomes visibly impossible.

Chapter III in twelve questions

A rational number is:

any decimal

$\frac{p}{q}$ with $q \neq 0$

an integer

a fraction below 1

Is -4 rational?

yes

no

sometimes

undefined

3 8 as a decimal:

0.38

0.375

0.83

0.125

A fraction terminates when the denominator has only:

2 and 3

2 and 5

primes

odd factors

$\frac{2}{7} + \frac{3}{7}$ equals:

$$\frac{5}{14}$$

$$\frac{5}{7}$$

$$\frac{6}{7}$$

$$\frac{6}{49}$$

For unlike denominators, use:

the product

the LCM

the HCF

the sum

$-2\frac{1}{3} + 1\frac{2}{3}$ equals:

$$\frac{1}{6}$$

$$-\frac{1}{6}$$

$$\frac{5}{6}$$

$$-\frac{5}{6}$$

$a - b$ equals:

$$a + b$$

$$a + (-b)$$

$$b - a$$

$$-a + b$$

$-2\frac{1}{3}$ as an improper fraction:

$$-\frac{7}{3}$$

$$-\frac{5}{3}$$

$$\frac{7}{3}$$

$$-\frac{1}{3}$$

$4\frac{1}{5}$ with one whole borrowed:

$$3\frac{6}{5}$$

$$3\frac{11}{5}$$

$$3\frac{1}{5}$$

$$5\frac{1}{5}$$

For a chain, take the LCM of:

the first two

all of them

the largest

the smallest

$x + a = b$ gives:

$$x = a + b$$

$$x = b - a$$

$$x = a - b$$

$$x = ab$$

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O'ZBEKCHA	РУССКИЙ
Control work	Nazorat ishi	Контрольная работа
Rational number	Ratsional son	Рациональное число
Common denominator	Umumiy maxraj	Общий знаменатель
Mixed number	Aralash son	Смешанное число
Recurring decimal	Davriy o'nli kasr	Периодическая дробь
Simplest form	Eng sodda ko'rinish	Несократимый вид
Equation	Tenglama	Уравнение
Diagnosis	Tashxis	Диагностика

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check**Check yourself**

Q1 loses a mark most often for:

a wrong sign

not simplifying

a decimal

no units

Q2 must show:

a fraction

the recurrence

a percentage

rounding

Q3 begins with:

the numerators

the LCM

the answer

a sketch

In Q5, converting to improper fractions avoids:

the LCM

borrowing

signs

checking

Q6 is solved by:

adding $\frac{2}{3}$

subtracting $\frac{2}{3}$

multiplying

dividing

Work on the mistakes means:

fix the wrong line

rewrite the solution

copy the answer

skip it

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up

7 PROBLEMS

1. -2.4 as a fraction

ANSWER $-\frac{12}{5}$

2. $3\frac{1}{4}$ as a fraction

ANSWER $\frac{13}{4}$

3. $\frac{7}{8}$ as a decimal

ANSWER 0.875

4. $\frac{5}{12}$ as a decimal

ANSWER $0.41666\dots$

5. $\frac{2}{3} + \frac{3}{4}$

ANSWER $1\frac{5}{12}$

6. $-\frac{1}{2} + \frac{1}{5}$

ANSWER $-\frac{3}{10}$

7. $\frac{5}{6} - \frac{3}{8}$

ANSWER $\frac{11}{24}$

Medium · the standard question

7 PROBLEMS

1. $-\frac{2}{5} - \frac{1}{2}$

ANSWER $-\frac{9}{10}$

2. $3\frac{1}{3} - 1\frac{3}{4}$

ANSWER $1\frac{7}{12}$

3. Solve $x + \frac{2}{3} = \frac{1}{6}$

ANSWER $-\frac{1}{2}$

4. $\frac{1}{2} + \frac{1}{3} - \frac{1}{4}$

ANSWER $\frac{7}{12}$

5. $0.6 + \frac{1}{5}$

ANSWER $\frac{4}{5}$

6. Simplify $\frac{45}{75}$

ANSWER $\frac{3}{5}$

7. Solve $x - \frac{1}{4} = \frac{1}{2}$

ANSWER $\frac{3}{4}$

Hard · stretch and reason

7 PROBLEMS

1. $2\frac{1}{6} - 3\frac{1}{2}$

ANSWER $-1\frac{1}{3}$

2. $-\frac{3}{4} + \frac{5}{6} - \frac{1}{12}$

ANSWER 0

3. Which is larger, $\frac{7}{12}$ or 0.6 ?

ANSWER 0.6

4. Solve $\frac{5}{6} - x = -\frac{1}{3}$

ANSWER $\frac{7}{6}$

5. The mean of $\frac{1}{4}, \frac{1}{2}, \frac{3}{4}$

ANSWER $\frac{1}{2}$

6. A tank $\frac{2}{3}$ full loses $\frac{1}{4}$ of the tank

ANSWER $\frac{5}{12}$ full

7. Order $\frac{3}{5}, \frac{5}{8}, \frac{7}{12}$

ANSWER $\frac{7}{12} < \frac{3}{5} < \frac{5}{8}$

07 Homework

Homework — 5 tasks

Rewrite in full every question that lost a mark before the second quarter begins.

1. Write -1.8 and $2\frac{3}{5}$ as fractions.
2. Convert $\frac{9}{16}$ and $\frac{7}{15}$ to decimals.
3. Compute $\frac{3}{5} + \frac{1}{4}$ and $-\frac{2}{3} + \frac{1}{6}$.
4. Compute $4\frac{1}{4} - 2\frac{2}{3}$.
5. Solve $x + \frac{3}{5} = \frac{1}{10}$.